

Forecasting COVID-19 Incident Hospitalizations with Temporal Hierarchies

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Introduction

The COVID-19 pandemic has underscored how infectious disease outbreaks can strain public health, clinical, and economic institutions. This has highlighted the importance of infectious disease forecasting as a way for the scientific community to communicate information, both to policymakers and the general public, about future directions of the pandemic. Outbreaks of other transmissible diseases like dengue, ebola, SARS, and influenza have also seen substantial forecasting efforts to reduce their spread and wider impact, modelers making quantitative predictions about the future based on already-observed data. These predictions may pertain to a specific event, outcome, or trend related to one (or occasionally more) infectious disease (Lauer, 2020). In this way, decisions that balance prevention or mitigation with economic or societal loss can be made much more effectively (Smith, 2019). For example, healthcare staffing and allocation of limited resources may be determined by any forecasted trends. Targeted communication campaigns about social distancing or vaccination may also be scheduled ahead of any forecasted increases in order to be most effective. Similarly, lockdowns may be eased during a forecasted downward trend to help boost the economy.

COVID-19 forecasts (and many infectious disease forecasts in general) tend to be short-term, predicting up to no more than four weeks ahead. Further, this type of forecast is quantifiable and specific for a fixed period of time (e.g. one week ahead, two weeks ahead, etc.) and can therefore be evaluated by comparing the forecasts to the eventually observed data. Although longer-term COVID-19 scenario projections that predict further out into the future under specified hypothetical conditions are useful in making more lasting plans for combating infectious disease, only short-term COVID-19 forecasts are explored in this work.

Short-term COVID-19 forecasting has been heavily researched with efforts synthesized, collected, archived, and evaluated in a centralized repository called the US COVID-19 Forecast Hub [<https://covid19forecasthub.org/>]. COVID-19 forecasts have also been in the form of probabilistic forecasts rather than point forecasts. The use of probabilistic forecasts is a distinguishing characteristic of outbreak forecasting efforts, as this type of forecast provides more information. For example, a probabilistic forecast quantifies the the likelihood of a forecast's occurrence.

However, obtaining accurate forecasts is not an easy task. For example, the systems that govern the spread of infectious diseases tend to be non-stationary processes since they are partially dependent on human behavior. For COVID-19, this non-stationarity was caused by continually changing policies and behaviors due to school closures, lockdowns, masking, social distancing, and the appearance of new variants. This makes trends difficult to forecast. Not only that, any forecasts made were used to inform those changing policies, further complicating matters due to the forecast feedback loop, a paradox that occurs when infectious disease forecasts guide intervention or risk communication, and thus impacting the path of the epidemic (Cramer, 2021; Lauer, 2020).

Forecasting COVID-19 also sees the additional challenge of an emergent outbreak setting, in which data availability and quality is severely limited (Lauer, 2020). This was especially true at the beginning of the pandemic, but a lack of a national testing protocol and issues properly categorizing deaths has meant that

models do not have accurate data to train on. This means that incident hospitalizations have started to become a more relevant public health metric compared to incident cases and incident deaths, despite its fewer previous forecasting efforts, making it an appropriate forecast target to explore.

Recent research about COVID-19 forecasting has also shown that most models have trouble predicting change-points in which trends in the data change direction (that is, hospitalizations start to increase or decrease in a way that is different from the previous pattern). Not only that, despite a range in data sources and modeling techniques, many models do not assign probability to the potential of a large change in trend. This gap calls for the exploration and testing of new methods.

Temporal Hierarchical Forecasting (THieF) created by Athanasopoulos et al in 2017 is a novel method that combines hierarchical forecasting and temporal aggregation for time series. This approach consists of three main steps: creation of a temporal hierarchy through data aggregation, generation of forecasts at every level of the hierarchy using a classical statistical model, and reconciliation of those forecasts. The final step performs weighted forecast combination to not only achieve coherence but also improved accuracy (Athanasopoulos, 2017). Coherence describes when hierarchical forecasts predict the same trend, event, or outcome at every level; reconciliation is the process of making forecasts coherent. If forecasts are not coherent, it is unclear which level of forecasts should be trusted and chosen over other levels (Kourentzes, 2021).

In light of the above-mentioned challenges within infectious disease forecasting, THieF might have something to offer. First, the THieF methodology relies on a phenomenological model, and for short-term outbreak forecasting, despite the growing body of knowledge we have about the mechanisms underlying disease transmission, models that focus only on the observed data, without incorporating a mechanistic understanding of outbreaks, often show good performance. Second, to my knowledge, THieF has not been applied in outbreak forecasting previously, meaning that it might offer insight about effective modeling techniques, regardless of its performance. Third, THieF has shown significant performance gains in other settings, which may translate to infectious disease forecasting.

The THieF methodology was used to retrospectively create probabilistic forecasts for COVID-19 incident daily hospitalizations for the United States at the state and national level. Six models were created based on the THieF methodology using an ARIMA base forecaster, and then were compared against each other and a naive baseline model during the training phase to find the most accurate models. Success was demonstrated by a few models shown to have accuracy gains when compared to the baseline. All of these forecasts were implemented through R using the package THieF made available by Athanasopoulos, et. al (2017).

Methods

Data

The data used in this analysis is the HealthData.gov COVID-19 Reported Patient Impact and Hospital Capacity by State Timeseries which reports daily incident hospitalizations for all 50 states, several US territories, and the aggregated U.S. national level. The sources of this state-aggregated time series are HHS Tele-Tracking, state/territorial health department reports to HHS Protect, and (prior to July 15, 2020) National Healthcare Safety Network. The COVID-19 Forecast Hub [<https://covid19forecasthub.org/>] designated this data set as the official ground truth data for COVID-19 incident hospitalizations on December 1, 2020 for modeling teams to train with and for models to be compared against. The data are collected at a daily frequency but are reported weekly through the covidHubUtils R package. Reporting for hospitalizations was sporadic during the early stages of the pandemic and only became consistent for all locations in July 2020. However, once hospitalization reporting became required for Medicare reimbursement, hospitalization data became much more reliable than case and death data, which suffer from major underreporting issues.

Forecast quantiles

As specified by the COVID-19 Forecast Hub, the 23 quantiles $Q = \{.010, 0.025, .050, .100, \dots, .900, .950, .990\}$ are used to represent a single forecast’s probabilistic distribution with the median 0.50 quantile typically taken as the point forecast. However, instead of specifying a full probabilistic distribution, I narrow the scope to make interval forecasts whose bounds can be used to obtain the desired 23 quantiles instead. The following table summarizes the translation from prediction interval bounds to quantiles, where “L” indicates the lower bound of a quantile while “U” indicates the upper bound.

Table 1: Quantile-Interval

Quantile	.010	.025	.050	.100	.150	.200	.250	.300	.350	.400	.450	.500	...	.975	.990
Interval	98L	95L	90L	80L	70L	60L	50L	40L	30L	20L	10L	Median	...	95U	98U

Forecast locations

Forecasts are made for all 50 states, Washington DC, Puerto Rico, and the US as a whole with a forecast horizon up to 84-day ahead incident hospitalizations, depending on the forecasting model (see Model Specifications below). Other US territories such as American Samoa and Guam are not included due to their very low or zero COVID-19 incident hospitalizations over the course of the pandemic. Forecasts that predicted single digit or even zero daily hospitalizations for these locations were often very accurate, meaning that these locations do not add much to understanding of differences in forecast skill between different modeling approaches (Cramer, 2021).

Date range Of analysis

The period of analysis was initially defined to be 462 days (66 weeks) long, starting on Monday, July 27th 2020 and ending Monday, November 1st 2021. This time period is split into a 343 day (49 week) training phase that ends on Monday, July 5th 2021 and a 119 day (17 week) testing phase that ends on Monday, November 1st 2021. The division is based on including two waves (the winter 2020-21 and the Alpha variant) of the pandemic in the training phase and one (Delta variant) in the testing phase. July 27th 2020 marks the beginning of when the forecast locations of interest had reliable truth data.

Only training phase analysis is included in this thesis to avoid preemptively looking at testing phase data; further work will contain testing phase analysis.

Training/validation and testing

The training phase is used to fit candidate models (see Model Specification subsection below) to the data, which are used to make forecasts between December 7th 2020 and July 5th 2021. (December 7th 2020 is the first Monday after the HealthData.gov incident hospitalizations dataset was designated the official ground truth data.) The training phase also serves as a validation period in which three models are chosen to move on to the test phase.

Two of the models will be the overall best based on forecast accuracy during the training phase while the remaining model will be the single best model in terms of forecast accuracy during the last 12 weeks of the training phase. This selection rule is to account for any models that perform better with more data to train on, as they will have this greater amount of data to help inform their forecasts during the testing phase.

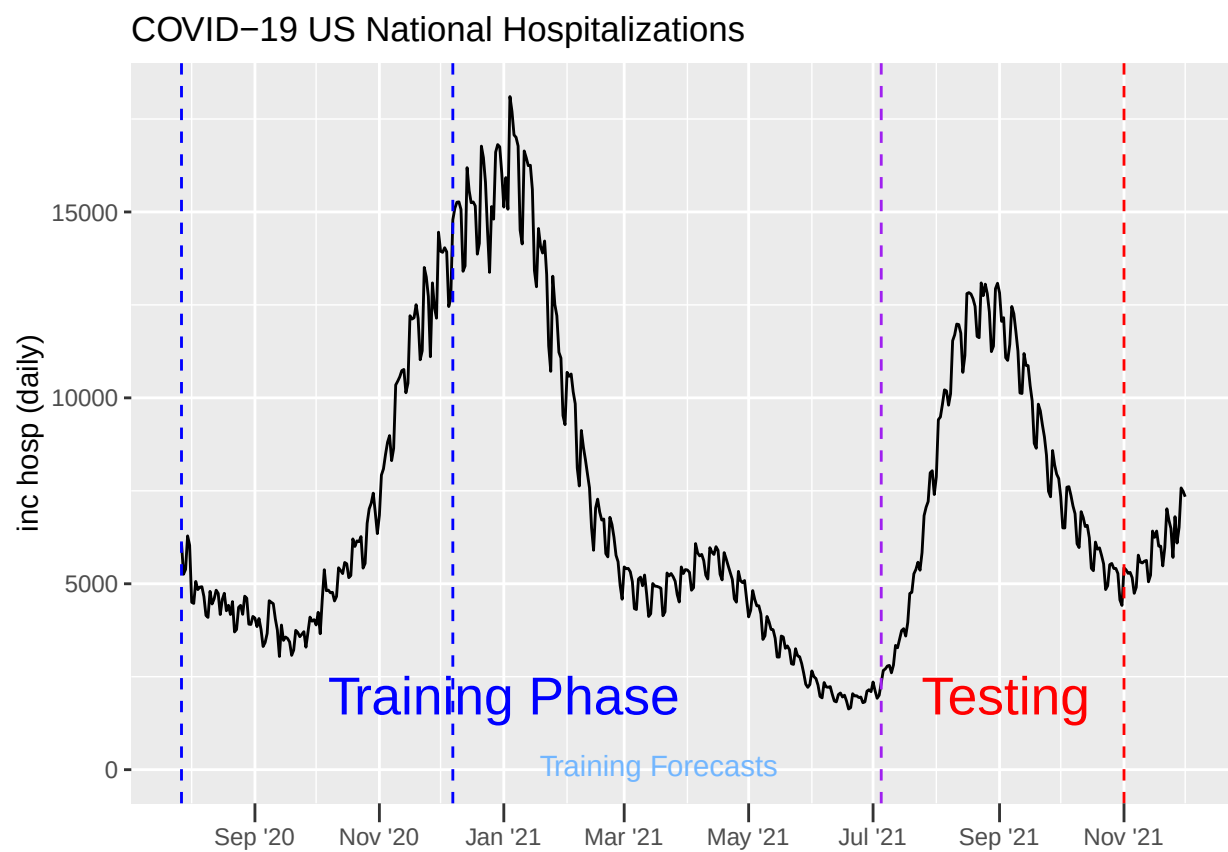


Figure 1: Period of Analysis Phases

Temporal hierarchical forecasting

Hierarchical forecasting has historically been utilized in business, economics, operations management, and related fields. This type of forecasting has seen useful advantages like producing coherent forecasts and enjoyed certain accuracy gains. Forecast combination, too, has been applied in a myriad of settings with favorable outcomes. The unique merging of these techniques in the THieF methodology has led to its promising accuracy improvements over more traditional hierarchical forecasting that does not involve forecasts combination.

The THieF approach has shown promising accuracy improvements over conventional methods due to its use of forecast combination. This forecast combination, while different from that of ensemble methods which combine forecasts from different, independent modeling frameworks, shares certain similarities which may indicate it could also show better performance. Research has also shown that classical statistical models often show good performance for short-term outbreak forecasting, suggesting another aspect of this methodology that may be beneficial.

Further, THieF’s technique of combining forecasts at every level means that this method strikes a balance between including key information from lower aggregation levels and detecting bigger picture trends more evidence at higher aggregation levels. In this way, THieF potentially could better pinpoint the true trend even amid noisy data and see less impact on its forecasts from outliers at lower levels. These are particularly attractive qualities because infectious disease surveillance data is often filled with noise due to substantial reporting delays and data revisions.

As previously mentioned, THieF is a three-step method for forecasting time-series data using a time-based hierarchical structure. These steps (outlined in more detail below) consist of data aggregation, base forecast creation, and reconciliation of the base forecasts.

The first step in using the THieF methodology is to create the hierarchical structure. Time-series data of a particular time scale can be aggregated into the desired levels of the hierarchy, where the original time scale defines the lowest temporal level (Athanasopoulos, 2017). For example, a three-level hierarchy can be created from a quarterly time series in which the data is aggregated to semi-annual and annual levels. Although in this case all the levels feed neatly into each other, the only other restriction on the temporal hierarchy is that all of the levels must be factors of the top-level (Athanasopoulos, 2017). Returning to the example, by thinking of the aggregation levels in terms of months, it can be seen that all of the levels (3-month, 6-month, 12-month) are all factors of the topmost level. Other temporal scales (in terms of months) that could be added to the hierarchy could be any other factors of 12.

Next, a classic statistical modeling approach is used to make the base forecasts at every level of the hierarchy. Athanasopoulos et al (2017) test both ETS and ARIMA approaches, with ARIMA shown to be more accurate. Based on these results, I choose to only use ARIMA based forecasters for all of my models. The model is fit to each aggregation level separately, as if they were separate time series, meaning that the order and values of parameters are allowed to vary. Hence, at this stage the forecasts are usually not coherent (Athanasopoulos, 2017).

Finally, these forecasts are optimally combined to obtain the reconciled forecasts. A simplified weighted least squares (WLS) estimator is used for this process (Athanasopoulos, 2017). The series variance scaling WLS estimator is used for the reconciliation process to determine the appropriate weights for each level of base forecasts. Forecasts with less variance of “reconciliation error” (the difference between base forecasts and their expected value if they were reconciled) are favored in this process (Athanasopoulos, 2017).

For my work on forecasting COVID-19 incident hospitalizations, the 4-week set of time scales consists of 1-day, 7-day, 14-day, and 28-day. These can also be thought of as daily, weekly, 2-weekly, and 4-weekly, and will be referred to as such in the rest of this thesis for convenience (Figure X). Each aggregation level can be thought of as a univariate time series from which forecasts can be made. Next, a classical statistical model is used to make these base forecasts. These consist of forecasts of up to 84-day ahead incident hospitalizations, depending on the forecasting model. I use the automatically generated prediction intervals from the `forecast` package to obtain the values for the 23 quantiles specified earlier (see Table 1 above). Then I optimally combine these base forecasts to obtain reconciled forecasts using the series variance scaling

estimator defined by Athanasopoulos, et. al (2017). This estimator determines the amount of weight to give the forecasts of each aggregation level when they are combined and reconciled based on the variance of their estimated in-sample forecast errors.

It is important to note that the original THieF method defined by Athanasopoulos, et. al (2017) is only defined for point forecasting. Further work on this methodology was developed by Taieb and Gamakumara to expand THieF to probabilistic forecasting but were deemed beyond the scope of this thesis due to the complexity of these developments. Instead, prediction intervals were used in place of fully specifying a probabilistic distribution (see Forecast Quantiles). Additional discussion can be found in Conclusions/Discussion.

Model specifications

During the training/validation phase, I compare several different specifications for the candidate models. Various combinations of modeling frameworks and timescale sets for the temporal hierarchies are used to generate the candidate models. More specifically, I test three hierarchical structures and two data transformations, resulting in six total candidate models.

Three different sets of time scales for the aggregation levels within the temporal hierarchy are explored in this work. The THieF_4wk set of time scales consist of daily, weekly, 2-weekly, and 4-weekly. The THieF_8wk set is made up of the THieF_4wk set plus an additional 8-weekly level that serves as the new topmost level. The THieF_12wk set adds 3-weekly, 6-weekly, and (topmost) 12-weekly time scales together in addition to the covidTHieF_4wk set. Note that the lower levels of each hierarchy are all multiples of that hierarchy's top level, an essential component to the THieF method.

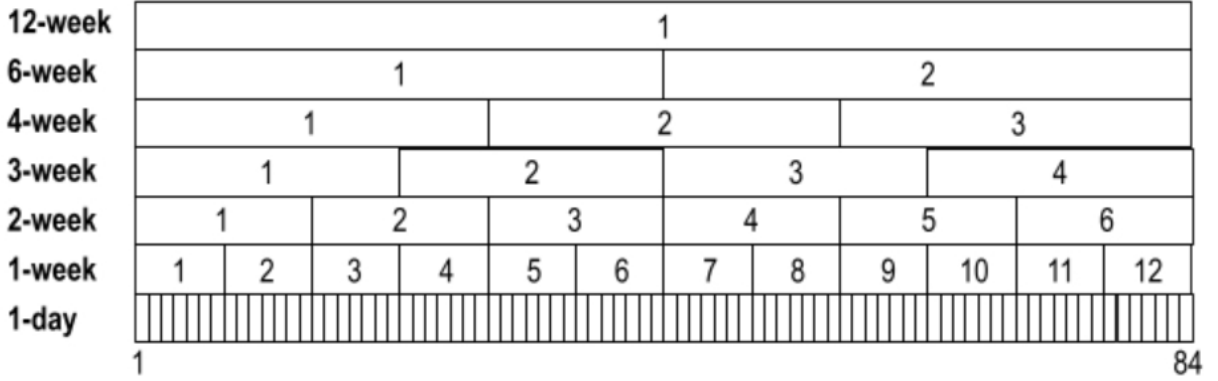
4-week Set

4-week	1																											
2-week	1														2													
1-week	1							2							3							4						
1-day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

8-week Set

8-week	1																																																							
4-week	1																												2																											
2-week	1														2														3														4													
1-week	1							2							3							4							5							6							7							8						
1-day																																																								
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12-week Set



Additionally, each of these hierarchy schemes is explored with both no data transformation and a variance-stabilizing fourth-root data transformation. The latter transformation is chosen based on prior work by the Reich Lab at UMass Amherst that has shown accuracy improvements when using a fourth root transformation for forecasting COVID-19.

Metrics and evaluation

The evaluation is conducted through several different aggregation schemes: horizon, target end date, location, pandemic wave (when applicable), and overall. Although incident hospitalizations are observed and forecasted at a daily level, I choose to evaluate the forecasts on a weekly basis. This is the standard in the monthly evaluation reports from the COVID-19 Forecast Hub and is computed by averaging the values for the metric of choice across a single week (Tuesday through Monday) to obtain a single value for that week. The aggregation scheme based on horizon, location, and pandemic wave will also be calculated similarly by averaging values by week.

I use the same three metrics as the COVID-19 Forecast Hub: MAE for point forecasts and WIS and PI coverage for probabilistic forecasts. Relative versions of MAE and WIS are also used when applicable, and are calculated as $MAE_{modelm}/MAE_{baseline}$ and $WIS_{modelm}/WIS_{baseline}$, respectively. Equations for these metrics are taken from Cramer, et. al (2021) and are shown below:

1. $MAE = \frac{1}{N} \sum_{i=1}^N |y_i - y_t|$
2. $WIS_{\alpha_0, K}(F, z_t) = \frac{1}{K+1/2} (w_0 \cdot |z_t - m| + \sum_{k=1}^K w_k \cdot IS_{\alpha_k}(F, z_t))$
3. $PICoverage = \frac{1}{N} 1(l_t^\alpha \leq z_t \leq u_t^\alpha)$

where $IS_{\alpha_k}(F, z_t) = (u_t - l_t) + \frac{2}{\alpha} \cdot (l_t - z_t) \cdot 1(z_t < l_t) + \frac{2}{\alpha} \cdot (z_t - u_t) \cdot 1(z_t > u_t)$.

This evaluation was intended to be performed both during the training/validation and testing phases separately, but only the training/validation phase analysis is provided here. The training and testing phases are separated based on a forecast date of July 5th 2021. However, the final forecasts for the training period can only be evaluated using data within the date range of the training phase, meaning that forecasts for that period are truncated such that the target end date must also be within the training phase as well. The training/validation evaluation compares the six models against each other as well as the naive baseline model created by the COVID-19 Forecast Hub, and this evaluation determines which three models to move on to the testing phase. (The testing phase will be conducted in further work and will involve a more in-depth accuracy evaluation and comparison with other models that forecast COVID-19 incident hospitalizations and submit to the COVID-19 Forecast Hub.)

No statistical tests are performed to assess the significance of differences in performance between models. As mentioned in the literature review, there are issues when forecasting time series that prevent the use

of such tests. Namely, scores are usually highly correlated across many dimensions, like horizons, forecast dates, locations, and more. Standard inference tends to make assumptions that the data/forecasts are i.i.d (independent and identically distributed), but these assumptions are invalid in such a case.

Results

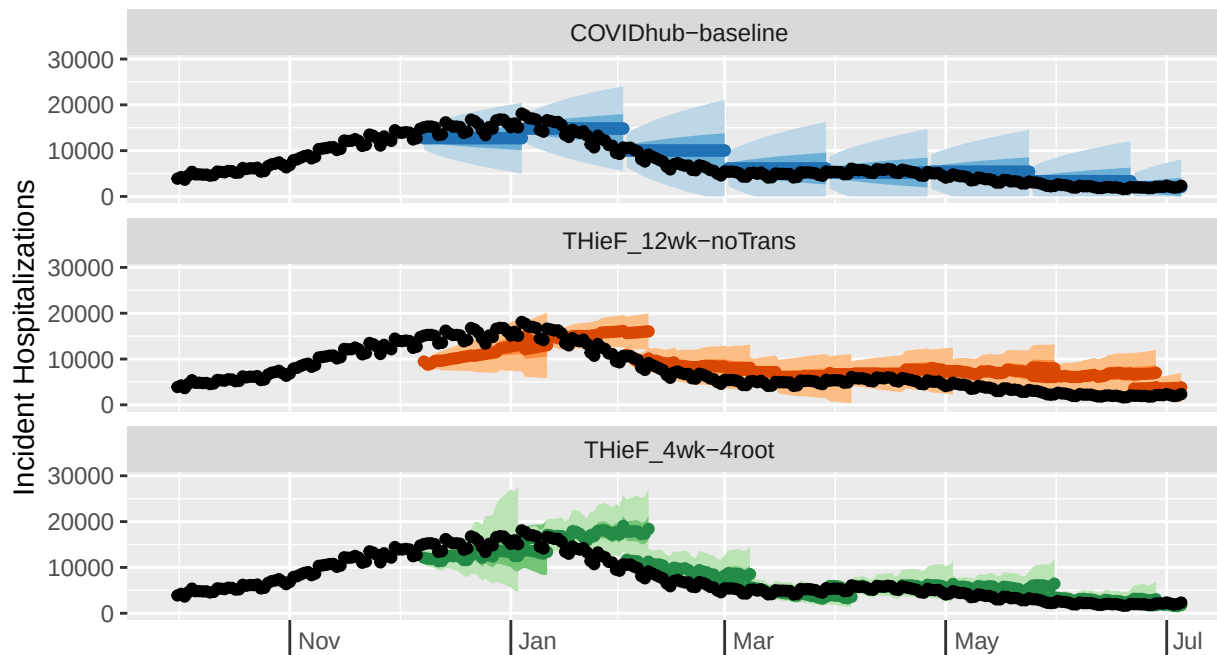
For every evaluation following the overall model performance, I evaluate performance at each horizon week separately, as performance often differs. Further, we also evaluate the US National level separately from the state/territory level given the huge differences and scale between values as well as the fact that the former is a sum of the latter. The performance for the state/territory level is averaged across locations.

Overall model performance

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): United States

Selected forecast date(s): 2020-12-07, 2021-01-04, 2021-02-01, 2021-03-01, 2021-04-01, 2021-05-01, 2021-06-01, 2021-07-01

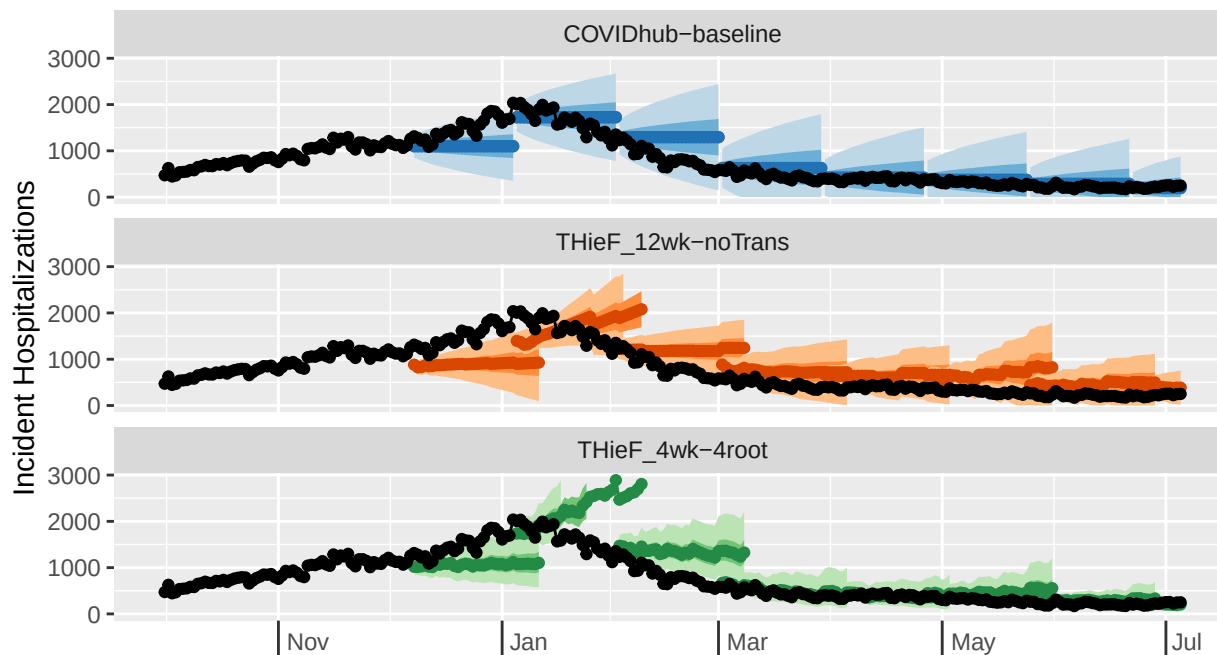


source: HealthData (observed data), THieF_4wk-4root, COVIDhub-baseline, THieF_12wk-noTrans (forecasts)

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Texas

Selected forecast date(s): 2020-12-07, 2021-01-04, 2021-02-01, 2021-03-01, 2021-04-01, 2021-05-01, 2021-06-01, 2021-07-01

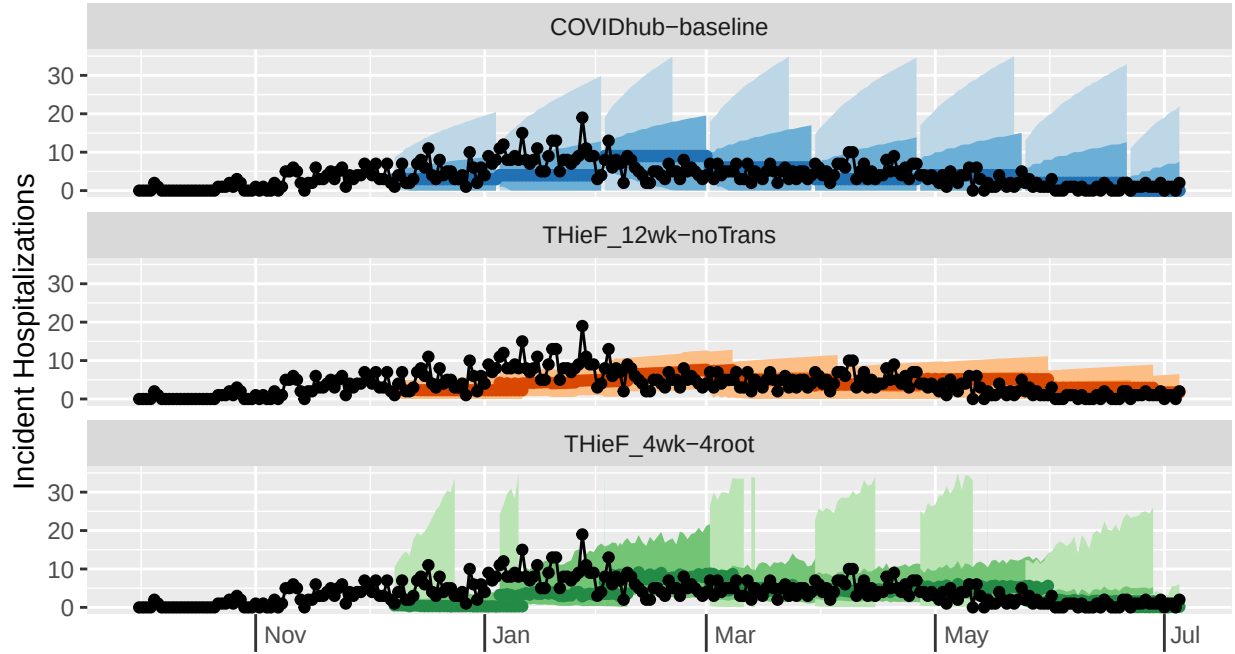


source: HealthData (observed data), THieF_4wk-4root, COVIDhub-baseline, THieF_12wk-noTrans (forecasts)

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Vermont

Selected forecast date(s): 2020-12-07, 2021-01-04, 2021-02-01, 2021-03-01, 2021-04-01



source: HealthData (observed data), THieF_4wk-4root, COVIDhub-baseline, THieF_12wk-noTrans (forecasts)

For the US National level (Table 2), the two THieF_4wk models take the lead for forecast accuracy and show better performance than the naive baseline for all metrics except 95% PI coverage. In terms of WIS, the THieF_4wk-4root model performs the best with a relative WIS value of 0.924 and a relative MAE value of 0.938, meaning it had 7.6% and 6.2% improvements over the baseline. This model also had 30.3% and 81.0% coverage rates for 50% and 95% prediction intervals, respectively, the former slightly closer to the nominal coverage rate than the baseline. In terms of 50% coverage rate, THieF_4wk-noTrans performed the best with a 50% coverage rate of 40.7% and a 95% coverage rate of 83.9%, the former a decent amount closer to the nominal coverage rate than the baseline. This model also had a 0.965 relative WIS and 0.937 relative MAE, or 3.5% and 6.3% improvements over the baseline. The other models show poorer performance compared to the baseline, with the THieF_8wk models performing about equally or worse than the baseline but the THieF_12wk models performing substantially worse than the baseline.

For the aggregated states level (Table 3), there is much greater similarity between the THieF_4wk and THieF_8wk models in terms of all metrics except 95% PI coverage. These models show more substantial improvements over the baseline compared to that for the US National level for all metrics except for relative MAE, for which the best models perform slightly worse than the baseline. The THieF_4wk-4root model performs the best for all metrics except for 95% PI coverage, although it is a very close third best in terms of being closest to the nominal coverage rate. This model has a relative WIS of 0.873 (12.7% improvement over the baseline), 1.007 relative MAE (0.7% worse than the baseline), 47.8% 50% PI coverage (the baseline has an 88.8% 50% coverage rate) and 91.6% 95% PI coverage (the baseline has a 100% 95% coverage rate). The THieF_8wk-4root model has similar relative WIS and relative MAE values (but worse coverage rates) while the THieF_8wk-noTrans model has similar coverage rates (but worse relative WIS and relative MAE). The THieF_12wk show certain improvements in coverage rates but do not show the same number of improvements over the baseline as the other for models.

Table 2: Overall model performance for the US national level

model	wis	mae	cov_50	cov_95	rwis	rmae
THieF_4wk-4root	1153.016	1748.479	0.303	0.810	0.924	0.938
THieF_4wk-noTrans	1204.261	1747.678	0.407	0.839	0.965	0.937
THieF_8wk-4root	1243.979	1934.376	0.229	0.690	0.997	1.038
COVIDhub-baseline	1248.094	1864.390	0.754	0.997	1.000	1.000
THieF_8wk-noTrans	1325.070	2054.114	0.329	0.819	1.062	1.102
THieF_12wk-4root	1546.332	2332.263	0.190	0.602	1.239	1.251
THieF_12wk-noTrans	1629.842	2555.703	0.169	0.747	1.306	1.371

Table 3: Overall model performance averaged over the states level

model	wis	mae	cov_50	cov_95	rwis	rmae
THieF_4wk-4root	30.763	46.968	0.478	0.916	0.873	1.007
THieF_8wk-4root	31.256	48.094	0.418	0.896	0.887	1.032
THieF_4wk-noTrans	32.115	47.769	0.577	0.915	0.912	1.025
THieF_8wk-noTrans	32.388	49.002	0.530	0.918	0.920	1.051
COVIDhub-baseline	35.222	46.620	0.888	0.991	1.000	1.000
THieF_12wk-noTrans	38.236	60.534	0.385	0.916	1.086	1.298
THieF_12wk-4root	56.122	54.990	0.373	0.856	1.593	1.180

Model performance by horizon week

Similar patterns in model accuracy are seen when forecasts are aggregated by horizon week for the US National level (see Supplemental Table 1). Although only the poor-performing models showed worse WIS and MAE values at a 1-week ahead horizon, all models had worse relative performance compared to the baseline at the same 1-week ahead horizon (see Figure 3). This is likely due to the models being overconfident at this horizon, as evidenced by their low PI coverage rates (see Supplemental Table 1). The THieF_4wk models once again generally show the best performance of the six models, especially at smaller horizons with THieF_8wk models performing similarly or better at a 4 week ahead horizon for certain metrics. The THieF_4wk-4root model performs better at smaller horizons while the THieF_4wk-noTrans model generally performs better at longer horizons. We see improved performance over the baseline for different metrics at different horizons for the THieF_4wk and THieF_8wk models, although no improvement is shown for 95% PI coverage for any of the models. In fact, all of the models show poor 95% PI coverage compared to the baseline.

Likewise for the aggregated states level, similar trends in model accuracy are seen in terms of model ranking when forecasts are stratified by horizon week compared to when there is no such stratification (see Supplemental Table 2). The greatest improvements over the baseline are seen in coverage rates, especially the 95% PI coverage, although the 50% PI coverage shows every model beating the baseline in terms of coverage rates closer to the nominal rate. This coverage rate improvement over the baseline at the states level is striking when compared to the US National level. On the other hand, only the THieF_8wk models show improved performance at the state level for every metric over the US national level while the THieF_4wk-4root model only shows improve performance at the state level for relative WIS. However, that all four THieF_4wk and covidTHieF_8wk models show improvement over the baseline for their relative WIS values for 2, 3, and 4 week ahead horizons.

Average WIS by horizon week and model

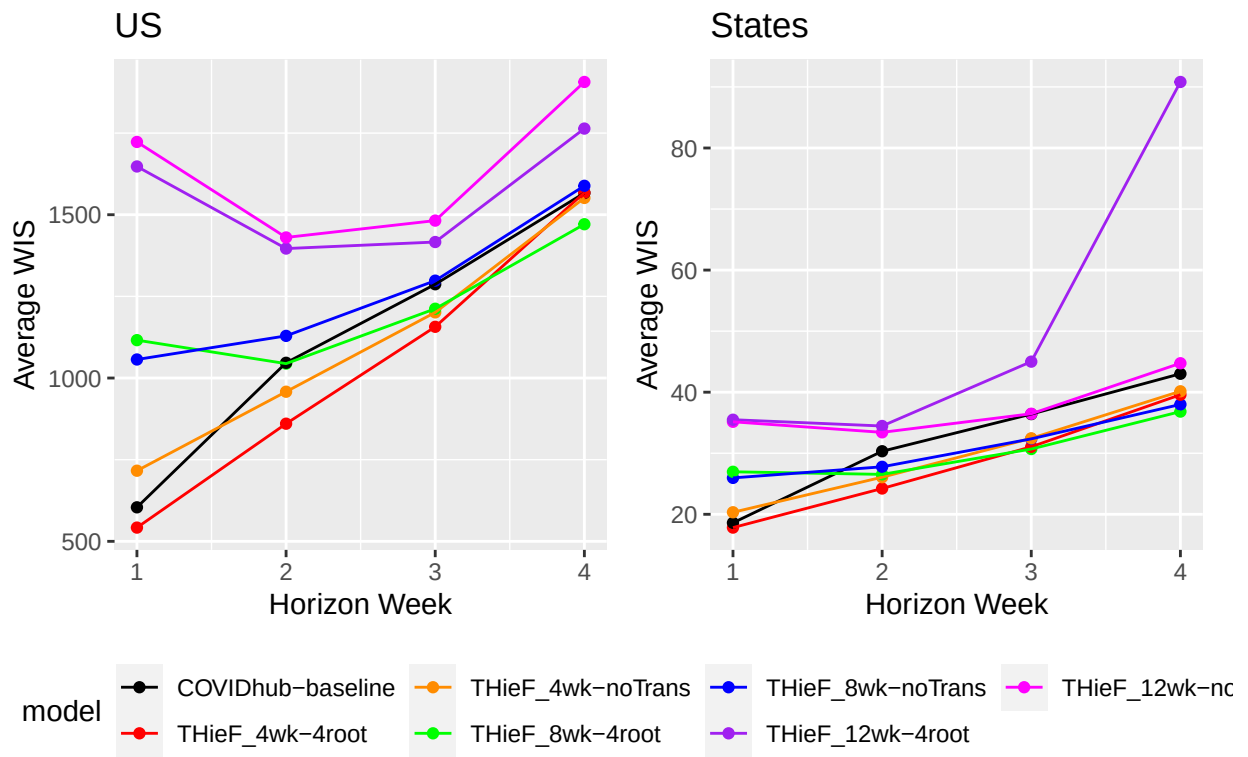


Figure 2: Average WIS by horizon week and model

Model performance by wave

For the US National level (see Supplemental Table 3 for full results), both waves yet again saw the THieF_12wk models performing worse at a 1-week ahead horizon compared to longer horizons while all models performed worse relative to the baseline at the same 1-week ahead horizon due to overconfidence. This is shown in Figure 4. Notably, most models performed better during the alpha wave in terms of every metric except relative MAE, likely due to the fact that the number of hospitalizations were smaller and thus had less variance. Once again, there tended to be worse coverage rates for the 95% PI compared to the 50% PI. During the alpha wave, the THieF_4wk_4root model showed the most improvements over the baseline in terms of relative WIS and MAE at every horizon, with between nearly 20% and 38% improvements for WIS and the third best coverage rates with some improvements over the baseline. The THieF_4wk_noTrans and THieF_8wk_noTrans models were the two best performing for coverage rate, beating the baseline for every horizon at the 50% level and matching it at the 95% level. During the winter21 wave, the two THieF_4wk models performed similarly, each performing the best for different horizons and different metrics. Generally, the THieF_4wk_4root model performed better for shorter horizons while the THieF_4wk_noTrans model performed better for longer horizons. Interestingly, at longer horizons the covidTHieF_12wk_noTrans model achieved some of the best relative WIS and MAE values of the winter21 wave, being between 10% and 19% better than the baseline despite poor performance in other instances.

For the states level (see Supplemental Table 4 for full results), there is somewhat of a someone of a similar trend of the worst performance relative to the baseline at 1-week ahead, although it is much less pronounced than that of the US National level (see Figure 4). The 95% coverage rates are also less poor compared to the US National level. During the alpha wave, the THieF_4wk_4root model similarly shows the most improvements over the baseline in terms of relative WIS and MAE at every horizon, with between close to 30% and 45% improvements for WIS and the best coverage rates with large improvements over the baseline except for one week ahead 95% coverage which was equal to that of the baseline. The THieF_4wk_noTrans model didn't always have the best performance across the four metrics but it was one of the top performing models that had consistent improvements over the baseline. The THieF_8wk_noTrans models showed some very competitive coverage rates that improved over the baseline for 50% coverage at every horizon but also had some instances of performing worse than the baseline for 95% coverage. During the winter21 wave, the two THieF_4wk models performed more similarly, though the THieF_4wk_4root model performed slightly better overall. In addition, the THieF_8wk models showed smaller relative WIS and relative MAE values compared to the THieF_4wk models performed for longer horizons. Interestingly, at longer horizons the THieF_12wk_noTrans model also achieved some of the best relative WIS and MAE values of the winter21 wave, being up to just over 10% better than the baseline despite poor performance in other instances.

Model Performance by Week

For the US National level, except for some poor performance at the start, the THieF_4wk models show lower or comparable WIS to the baseline across most of the training period with a forecast horizon of one week ahead. The THieF_8wk models show mostly comparable WIS to the baseline while the THieF_12wk models show mostly higher WIS than the baseline for a one week ahead horizon during the entire training period. On the other hand, for a four week ahead horizon the THieF_4wk models show only comparable WIS to the baseline not during change points. We see a very similar graph shape to that of the actual values for incident hospitalizations. Meanwhile, for four week ahead horizons the THieF_12wk models show comparable WIS to the baseline not during phases of low incidence, often showing comparable or even better performance compared to the baseline during change points. Once again, the THieF_8wk models show intermediate performance between the other two groups of models.

Very similar patterns are observed for one week ahead and four week ahead US National level forecasts in terms of MAE, although the best models (the two THieF_4wk ones) only tend to perform about comparably to the baseline, unlike for WIS. The weekly 95% PI coverage rate plots show more stable coverage rates at the 4-week ahead horizon compared to that out the 1-week ahead horizon. Except for the baseline, most models show very variable coverage up until around March 2021, some weeks hitting full coverage while

Average WIS by wave and model

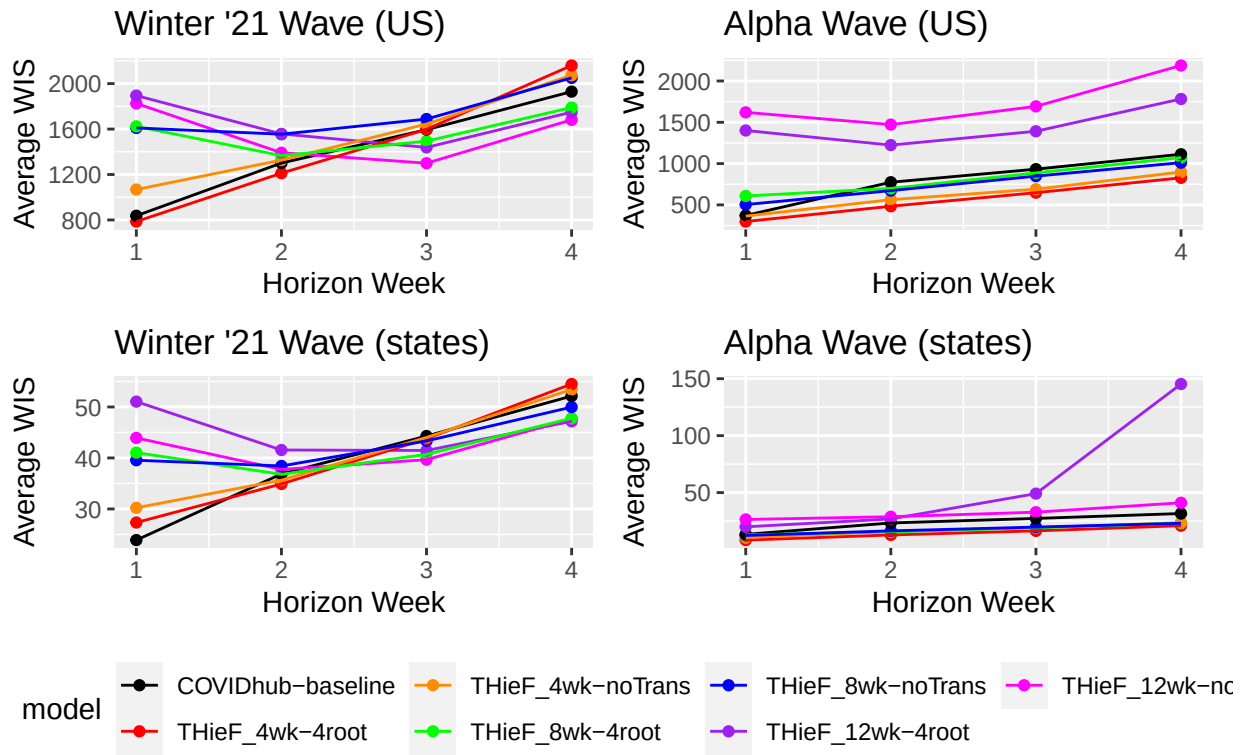


Figure 3: Average WIS by wave and model

others falling short for most forecast days. Similarly, most models do not show good coverage rates at a 4-week ahead horizon until that March 2021 cut off date. For both 1-week ahead and 4-week ahead horizons, the models without a data transformation tend to see consistently high or even 100% coverage for the 95% PI while the models with a fourth root transformation do not meet nominal coverage rates during certain periods. These do not quite line up with change points seen in incident hospitalizations.

For the states level aggregation, the same patterns are observed as for the US National level for one week ahead and four week ahead WIS and MAE, with strangely large four week ahead WIS values for the THieF_12wk_4root model. The weekly 95% PI coverage rate plots are more informative for the states level since they are an average across 52 locations, and they look like upside down versions of the weekly WIS plots. For the one week ahead horizon, the THieF_4wk models and the THieF_8wk_noTrans model show improved and fairly stable coverage starting in March 2021, likely due to smaller variance in incident values. For four week ahead horizon, all of the models without a data transformation seem to do the best starting from March 2021 with decent performance from THieF_4wk-4root and THieF_8wk-4root after the downward winter21 change point.

Average h-week ahead metric by model (US)

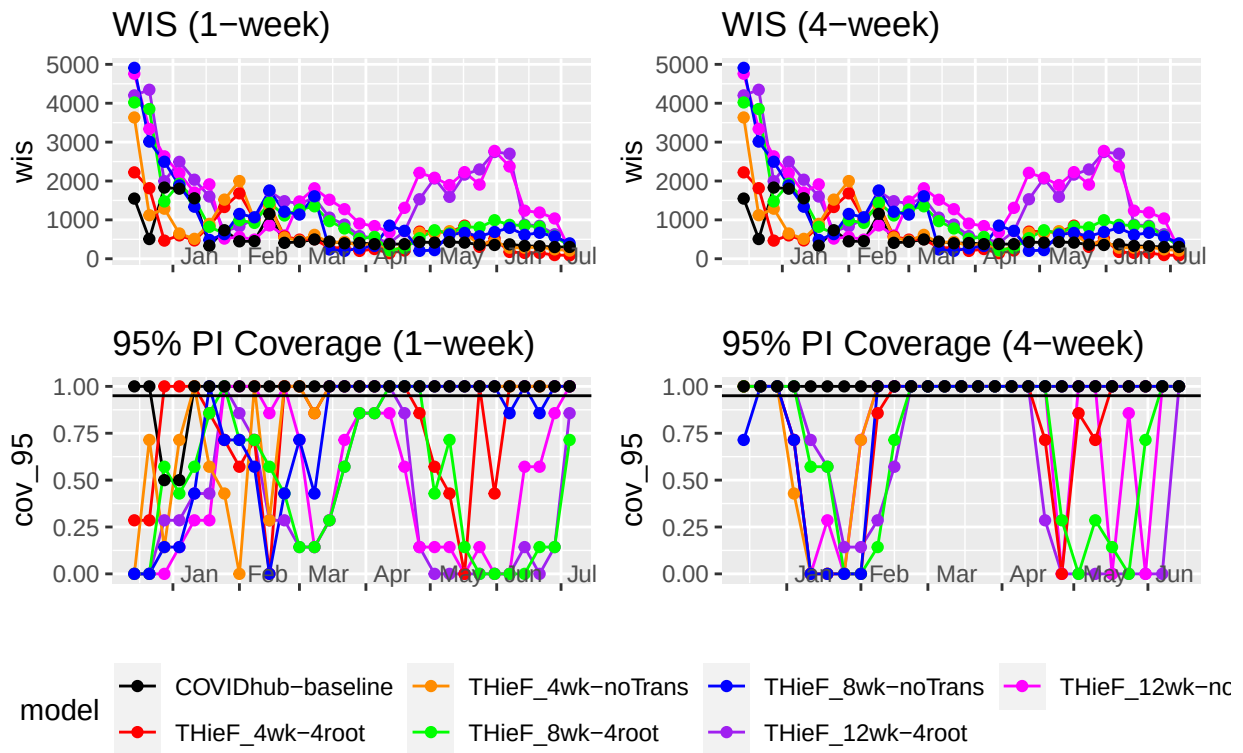


Figure 4: Average h-week ahead metric by model (US)

Conclusions/Discussion

Even as public health mandates lift and discussions of endemic COVID-19 become increasingly common, forecasting remains an important mode through which to gain information about COVID-19. Endemic diseases still see forecasting efforts due to their impact on hospitals, and these decisions can also impact the economy, especially through a loss of worker productivity. The path of COVID-19 also still sees shifting trends

Average h-week ahead metric by model (states)

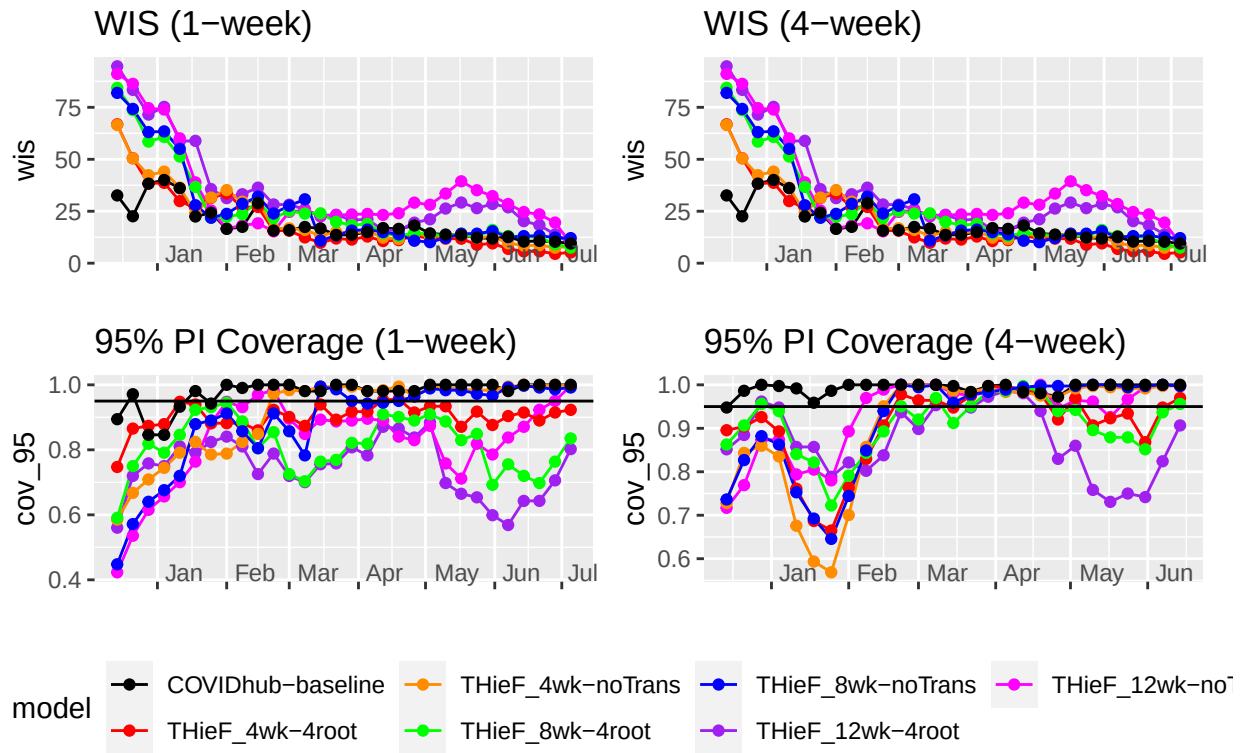


Figure 5: Average h-week ahead metric by model (states)

with each new variant, some of which could necessitate reinstating certain public health mandates. This paper provides an initial application of the THieF forecasting method to COVID-19 incident hospitalizations in the US from July 2020 to July 2021 through the creation and comparison of six distinct forecast combination approaches.

The THieF_4wk_4root model emerges as the overall best performing model across all evaluations regardless of stratification. This model consistently performs better or equal to the baseline for most metrics, horizons, and forecast dates at both the US National level as well as the state level. The THieF_4wk_noTrans model is the second best performing model of the THieF models overall and shows slightly better coverage rates at the US National level than that of THieF_4wk_4root. This model also shows consistently good performance compared to the baseline, although its the majority of its improvements are slightly smaller than that of THieF_4wk_4root. These two models show the most gains over the baseline in terms of WIS and 50% PI calibration, as can be seen in Figure [FIX_ME] which displays the generally much narrower prediction intervals of the THieF models compared to those of the baseline.

For the most part, it seems that the additions of the 8 week and 12 week aggregation levels did not serve to improve model accuracy. Since only 1 to 28 day ahead forecasts were evaluated, the higher levels may not have provided any additional insights about trends for these short-term horizons. However, the two models that included the 12 week hierarchy showed improved WIS and MAE over the baseline during change points at longer horizons, two settings that the two THieF_4wk models struggled to predict accurately, like many other existing forecasting models for COVID-19. Perhaps these two THieF_12wk models might offer some insights with further exploration.

While these results are promising, they are only initial results from the training phase. The performance of the subset of models that are passed onto the testing phase is still yet to be calculated, meaning the drawing of any definitive conclusions must be put on hold. Further investigation into model performance stratified by location would also serve to provide useful information to help determine which models perform the best under different circumstances, as well as an exploration of the prediction accuracy of a simple ARIMA model without the use of a temporal hierarchy.

In addition, there remain certain limitations of the THieF methodology in practice that may restrict its utility until further developments are made. The inherent structure of the temporal hierarchy only allows aggregation levels whose time scales are factors of the highest level’s time scale. This means that seasonality at any excluded timescales might be missed due to their exclusion, which is especially problematic when only limited data is available. Moreover, at the present, THieF lacks a simple probabilistic implementation, making it difficult to apply to areas in which probabilistic forecasting is the widely accepted standard. Future work points to applying the existing probabilistic implementation of THieF to forecasting COVID-19 incident hospitalizations. This implementation would allow for generation of a complete probabilistic distribution instead of just quantiles, which could prove to be useful. Still, the current work serves as indication that the THieF methodology performs well in an infectious disease setting, even amid challenges like system complexity, data sparsity/quality, the forecasting feedback loop, and warrants further exploration.

Supplemental Figures and Tables

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
THieF_4wk-4root	1	542.076	836.752	0.117	0.800	0.897	1.081
COVIDhub-baseline	1	604.156	773.900	0.800	0.967	1.000	1.000
THieF_4wk-noTrans	1	716.236	1063.424	0.083	0.783	1.186	1.374
THieF_8wk-noTrans	1	1056.776	1442.944	0.200	0.700	1.749	1.865
THieF_8wk-4root	1	1116.034	1512.255	0.067	0.333	1.847	1.954
THieF_12wk-4root	1	1647.772	2105.831	0.050	0.250	2.727	2.721
THieF_12wk-noTrans	1	1722.915	2232.146	0.083	0.317	2.852	2.884
THieF_4wk-4root	2	860.165	1323.950	0.393	0.841	0.822	0.856

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
THieF_4wk-noTrans	2	958.092	1359.289	0.476	0.828	0.915	0.879
THieF_8wk-4root	2	1044.122	1653.197	0.234	0.759	0.998	1.069
COVIDhub-baseline	2	1046.662	1546.717	0.786	1.000	1.000	1.000
THieF_8wk-noTrans	2	1129.085	1778.506	0.324	0.814	1.079	1.150
THieF_12wk-4root	2	1396.428	2135.817	0.145	0.628	1.334	1.381
THieF_12wk-noTrans	2	1430.444	2300.854	0.097	0.766	1.367	1.488
THieF_4wk-4root	3	1156.882	1765.098	0.337	0.827	0.899	0.914
THieF_4wk-noTrans	3	1200.986	1718.652	0.490	0.847	0.933	0.890
THieF_8wk-4root	3	1212.164	1919.171	0.250	0.755	0.942	0.994
COVIDhub-baseline	3	1287.123	1930.434	0.755	1.000	1.000	1.000
THieF_8wk-noTrans	3	1298.125	2080.367	0.321	0.847	1.009	1.078
THieF_12wk-4root	3	1416.465	2197.313	0.224	0.709	1.100	1.138
THieF_12wk-noTrans	3	1482.037	2403.128	0.219	0.852	1.151	1.245
THieF_8wk-4root	4	1470.919	2299.869	0.254	0.683	0.939	0.964
THieF_4wk-noTrans	4	1551.446	2292.973	0.370	0.857	0.990	0.961
COVIDhub-baseline	4	1566.581	2385.804	0.714	1.000	1.000	1.000
THieF_4wk-4root	4	1567.629	2346.378	0.259	0.772	1.001	0.983
THieF_8wk-noTrans	4	1588.545	2432.356	0.381	0.831	1.014	1.020
THieF_12wk-4root	4	1763.813	2694.806	0.233	0.582	1.126	1.130
THieF_12wk-noTrans	4	1906.551	3012.165	0.201	0.762	1.217	1.263

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
THieF_4wk-4root	1	17.822	25.568	0.504	0.892	0.958	1.102
COVIDhub-baseline	1	18.603	23.209	0.837	0.976	1.000	1.000
THieF_4wk-noTrans	1	20.333	28.783	0.523	0.903	1.093	1.240
THieF_8wk-noTrans	1	25.963	35.412	0.464	0.862	1.396	1.526
THieF_8wk-4root	1	26.963	37.196	0.382	0.788	1.449	1.603
THieF_12wk-noTrans	1	35.154	47.842	0.315	0.762	1.890	2.061
THieF_12wk-4root	1	35.491	47.142	0.317	0.720	1.908	2.031
THieF_4wk-4root	2	24.224	37.394	0.486	0.928	0.799	0.957
THieF_4wk-noTrans	2	26.076	39.368	0.577	0.928	0.860	1.008
THieF_8wk-4root	2	26.540	41.036	0.414	0.902	0.875	1.051
THieF_8wk-noTrans	2	27.778	41.741	0.531	0.927	0.916	1.069
COVIDhub-baseline	2	30.328	39.058	0.896	0.994	1.000	1.000
THieF_12wk-noTrans	2	33.424	52.824	0.372	0.929	1.102	1.352
THieF_12wk-4root	2	34.463	49.263	0.360	0.853	1.136	1.261
THieF_8wk-4root	3	30.690	47.891	0.425	0.914	0.843	0.999
THieF_4wk-4root	3	31.057	47.771	0.480	0.923	0.853	0.997
THieF_8wk-noTrans	3	32.357	49.562	0.532	0.928	0.889	1.034
THieF_4wk-noTrans	3	32.443	48.423	0.590	0.918	0.891	1.010
COVIDhub-baseline	3	36.412	47.923	0.896	0.993	1.000	1.000
THieF_12wk-noTrans	3	36.466	58.694	0.408	0.940	1.002	1.225
THieF_12wk-4root	3	45.006	52.951	0.392	0.884	1.236	1.105
THieF_8wk-4root	4	36.822	57.179	0.425	0.908	0.856	0.977
THieF_8wk-noTrans	4	37.998	58.305	0.548	0.919	0.883	0.997
THieF_4wk-4root	4	39.583	60.273	0.462	0.906	0.920	1.030
THieF_4wk-noTrans	4	40.149	59.564	0.581	0.906	0.933	1.018
COVIDhub-baseline	4	43.018	58.502	0.891	0.992	1.000	1.000
THieF_12wk-noTrans	4	44.741	72.387	0.394	0.931	1.040	1.237
THieF_12wk-4root	4	90.817	63.989	0.380	0.873	2.111	1.094

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
alpha	THieF_4wk-4root	1	298.271	478.368	0.100	0.833	0.803	1.477
alpha	THieF_4wk-noTrans	1	364.629	648.507	0.100	1.000	0.981	2.002
alpha	COVIDhub-baseline	1	371.549	323.933	1.000	1.000	1.000	1.000
alpha	THieF_8wk-noTrans	1	503.722	841.296	0.233	0.933	1.356	2.597
alpha	THieF_8wk-4root	1	608.219	882.186	0.000	0.333	1.637	2.723
alpha	THieF_12wk-4root	1	1400.673	1766.602	0.000	0.200	3.770	5.454
alpha	THieF_12wk-noTrans	1	1620.340	2151.713	0.067	0.200	4.361	6.642
alpha	THieF_4wk-4root	2	483.611	757.371	0.400	0.914	0.625	0.820
alpha	THieF_4wk-noTrans	2	563.022	863.548	0.514	1.000	0.727	0.935
alpha	THieF_8wk-noTrans	2	672.864	1170.013	0.457	1.000	0.869	1.267
alpha	THieF_8wk-4root	2	699.167	1062.503	0.286	0.686	0.903	1.151
alpha	COVIDhub-baseline	2	774.310	923.143	0.986	1.000	1.000	1.000
alpha	THieF_12wk-4root	2	1224.000	1771.332	0.171	0.571	1.581	1.919
alpha	THieF_12wk-noTrans	2	1472.230	2394.273	0.100	0.743	1.901	2.594
alpha	THieF_4wk-4root	3	647.750	1044.362	0.319	0.901	0.695	0.893
alpha	THieF_4wk-noTrans	3	689.803	1088.364	0.549	1.000	0.740	0.931
alpha	THieF_8wk-noTrans	3	848.269	1468.658	0.484	1.000	0.910	1.256
alpha	THieF_8wk-4root	3	887.183	1363.911	0.220	0.670	0.951	1.166
alpha	COVIDhub-baseline	3	932.443	1169.429	0.989	1.000	1.000	1.000
alpha	THieF_12wk-4root	3	1391.170	2040.682	0.154	0.495	1.492	1.745
alpha	THieF_12wk-noTrans	3	1692.610	2754.007	0.154	0.890	1.815	2.355
alpha	THieF_4wk-4root	4	827.597	1349.386	0.226	0.857	0.743	0.904
alpha	THieF_4wk-noTrans	4	897.845	1508.112	0.440	1.000	0.807	1.011
alpha	THieF_8wk-noTrans	4	1011.064	1649.833	0.476	1.000	0.908	1.106
alpha	THieF_8wk-4root	4	1072.725	1672.564	0.190	0.619	0.964	1.121
alpha	COVIDhub-baseline	4	1113.129	1492.357	0.976	1.000	1.000	1.000
alpha	THieF_12wk-4root	4	1782.333	2639.003	0.119	0.357	1.601	1.768
alpha	THieF_12wk-noTrans	4	2187.866	3535.404	0.060	0.738	1.966	2.369
winter21	THieF_4wk-4root	1	785.881	1195.135	0.133	0.767	0.939	0.977
winter21	COVIDhub-baseline	1	836.762	1223.867	0.600	0.933	1.000	1.000
winter21	THieF_4wk-noTrans	1	1067.842	1478.342	0.067	0.567	1.276	1.208
winter21	THieF_8wk-noTrans	1	1609.830	2044.593	0.167	0.467	1.924	1.671
winter21	THieF_8wk-4root	1	1623.849	2142.325	0.133	0.333	1.941	1.750
winter21	THieF_12wk-noTrans	1	1825.491	2312.578	0.100	0.433	2.182	1.890
winter21	THieF_12wk-4root	1	1894.871	2445.060	0.100	0.300	2.265	1.998
winter21	THieF_4wk-4root	2	1211.615	1852.757	0.387	0.773	0.931	0.870
winter21	COVIDhub-baseline	2	1300.857	2128.720	0.600	1.000	1.000	1.000
winter21	THieF_4wk-noTrans	2	1326.824	1821.980	0.440	0.667	1.020	0.856
winter21	THieF_8wk-4root	2	1366.081	2204.511	0.187	0.827	1.050	1.036
winter21	THieF_12wk-noTrans	2	1391.443	2213.664	0.093	0.787	1.070	1.040
winter21	THieF_8wk-noTrans	2	1554.891	2346.433	0.200	0.640	1.195	1.102
winter21	THieF_12wk-4root	2	1557.360	2476.003	0.120	0.680	1.197	1.163
winter21	THieF_12wk-noTrans	3	1299.539	2099.034	0.276	0.819	0.815	0.810
winter21	THieF_12wk-4root	3	1438.387	2333.060	0.286	0.895	0.902	0.901
winter21	THieF_8wk-4root	3	1493.815	2400.396	0.276	0.829	0.937	0.927
winter21	COVIDhub-baseline	3	1594.512	2589.971	0.552	1.000	1.000	1.000
winter21	THieF_4wk-4root	3	1598.130	2389.736	0.352	0.762	1.002	0.923
winter21	THieF_4wk-noTrans	3	1644.012	2264.901	0.438	0.714	1.031	0.874
winter21	THieF_8wk-noTrans	3	1688.000	2610.515	0.181	0.714	1.059	1.008
winter21	THieF_12wk-noTrans	4	1681.499	2593.573	0.314	0.781	0.872	0.836
winter21	THieF_12wk-4root	4	1748.996	2739.448	0.324	0.762	0.907	0.884
winter21	THieF_8wk-4root	4	1789.474	2801.714	0.305	0.733	0.928	0.904

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
winter21	COVIDhub-baseline	4	1929.342	3100.562	0.505	1.000	1.000	1.000
winter21	THieF_8wk-noTrans	4	2050.529	3058.375	0.305	0.695	1.063	0.986
winter21	THieF_4wk-noTrans	4	2074.327	2920.862	0.314	0.743	1.075	0.942
winter21	THieF_4wk-4root	4	2159.655	3143.972	0.286	0.705	1.119	1.014

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
alpha	THieF_4wk-4root	1	8.306	12.731	0.512	0.906	0.624	0.961
alpha	THieF_4wk-noTrans	1	10.440	16.125	0.651	0.987	0.784	1.217
alpha	THieF_8wk-noTrans	1	12.375	19.697	0.563	0.966	0.930	1.486
alpha	THieF_8wk-4root	1	12.882	19.407	0.388	0.799	0.968	1.464
alpha	COVIDhub-baseline	1	13.308	13.254	0.947	0.994	1.000	1.000
alpha	THieF_12wk-4root	1	19.934	28.371	0.315	0.712	1.498	2.141
alpha	THieF_12wk-noTrans	1	26.375	39.816	0.281	0.789	1.982	3.004
alpha	THieF_4wk-4root	2	12.789	19.860	0.504	0.946	0.548	0.806
alpha	THieF_8wk-4root	2	15.551	24.841	0.411	0.907	0.667	1.008
alpha	THieF_4wk-noTrans	2	15.818	24.530	0.706	0.993	0.678	0.995
alpha	THieF_8wk-noTrans	2	16.369	24.147	0.702	0.989	0.702	0.980
alpha	COVIDhub-baseline	2	23.331	24.641	0.961	0.995	1.000	1.000
alpha	THieF_12wk-4root	2	26.861	35.382	0.346	0.840	1.151	1.436
alpha	THieF_12wk-noTrans	2	28.797	49.045	0.328	0.968	1.234	1.990
alpha	THieF_4wk-4root	3	16.394	25.759	0.502	0.955	0.600	0.877
alpha	THieF_4wk-noTrans	3	19.082	29.607	0.728	0.990	0.699	1.009
alpha	THieF_8wk-4root	3	19.134	30.761	0.411	0.929	0.700	1.048
alpha	THieF_8wk-noTrans	3	19.691	29.793	0.704	0.992	0.721	1.015
alpha	COVIDhub-baseline	3	27.315	29.355	0.961	0.994	1.000	1.000
alpha	THieF_12wk-noTrans	3	32.778	56.299	0.356	0.977	1.200	1.918
alpha	THieF_12wk-4root	3	49.094	40.979	0.361	0.868	1.797	1.396
alpha	THieF_4wk-4root	4	20.933	32.675	0.490	0.950	0.662	0.917
alpha	THieF_8wk-noTrans	4	23.027	35.651	0.716	0.996	0.728	1.000
alpha	THieF_8wk-4root	4	23.110	37.052	0.407	0.936	0.731	1.039
alpha	THieF_4wk-noTrans	4	23.545	36.957	0.715	0.989	0.745	1.037
alpha	COVIDhub-baseline	4	31.616	35.644	0.956	0.995	1.000	1.000
alpha	THieF_12wk-noTrans	4	41.065	70.591	0.333	0.977	1.299	1.980
alpha	THieF_12wk-4root	4	145.358	53.066	0.333	0.857	4.598	1.489
winter21	COVIDhub-baseline	1	23.898	33.164	0.728	0.958	1.000	1.000
winter21	THieF_4wk-4root	1	27.338	38.404	0.496	0.877	1.144	1.158
winter21	THieF_4wk-noTrans	1	30.227	41.441	0.395	0.819	1.265	1.250
winter21	THieF_8wk-noTrans	1	39.551	51.127	0.365	0.758	1.655	1.542
winter21	THieF_8wk-4root	1	41.044	54.985	0.377	0.778	1.717	1.658
winter21	THieF_12wk-noTrans	1	43.933	55.868	0.349	0.734	1.838	1.685
winter21	THieF_12wk-4root	1	51.049	65.913	0.320	0.728	2.136	1.987
winter21	THieF_4wk-4root	2	34.898	53.760	0.469	0.911	0.947	1.024
winter21	THieF_4wk-noTrans	2	35.651	53.217	0.457	0.866	0.967	1.013
winter21	THieF_8wk-4root	2	36.797	56.152	0.418	0.897	0.998	1.069
winter21	COVIDhub-baseline	2	36.858	52.513	0.835	0.993	1.000	1.000
winter21	THieF_12wk-noTrans	2	37.741	56.352	0.413	0.893	1.024	1.073
winter21	THieF_8wk-noTrans	2	38.427	58.162	0.372	0.869	1.043	1.108
winter21	THieF_12wk-4root	2	41.558	62.218	0.374	0.864	1.128	1.185
winter21	THieF_12wk-noTrans	3	39.663	60.769	0.454	0.908	0.895	0.949
winter21	THieF_8wk-4root	3	40.706	62.737	0.436	0.902	0.919	0.980

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
winter21	THieF_12wk-4root	3	41.464	63.327	0.420	0.898	0.936	0.989
winter21	THieF_8wk-noTrans	3	43.334	66.696	0.384	0.872	0.978	1.042
winter21	THieF_4wk-4root	3	43.765	66.848	0.460	0.896	0.988	1.044
winter21	THieF_4wk-noTrans	3	44.023	64.730	0.470	0.856	0.994	1.011
winter21	COVIDhub-baseline	3	44.295	64.016	0.839	0.992	1.000	1.000
winter21	THieF_12wk-4root	4	47.184	72.727	0.418	0.887	0.905	0.947
winter21	THieF_12wk-noTrans	4	47.682	73.824	0.442	0.895	0.914	0.961
winter21	THieF_8wk-4root	4	47.791	73.280	0.440	0.885	0.917	0.954
winter21	THieF_8wk-noTrans	4	49.975	76.428	0.414	0.858	0.958	0.995
winter21	COVIDhub-baseline	4	52.140	76.788	0.839	0.990	1.000	1.000
winter21	THieF_4wk-noTrans	4	53.433	77.649	0.474	0.839	1.025	1.011
winter21	THieF_4wk-4root	4	54.503	82.352	0.441	0.871	1.045	1.072